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Fire Prediction and Risk Identification With Interpretable Machine Learning

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ABSTRACT

Fire safety is a primary concern in safeguarding lives and property. However, it is challenging to predict fire incidents and identify potential influencing factors due to limitations of data, model accuracy and interpretability. This paper proposes a novel scheme designed to enhance predictive and explainable capabilities by integrating multi-source data, adaptive machine learning methods, and Shapley additive explanation (SHAP) tools for more effective and applicable fire safety management. The scheme shows satisfactory prediction results by leveraging the data from grid-style management systems and our proposed machine learning method with dynamic time warping distance-based time series clustering, significantly outperforming the methods merely based on time series modeling. Moreover, clustered features help to clarify the main influencing risk factors and provide clearer insights for model interpretability. With global SHAP, community clusters capturing community fire event frequency, as well as historical records on fire police rescue, smoke alarms, and fire alarms, are found to be significant risk factors among all the features over the whole communities and periods via the model interpretability analysis, implying that communities where fires used to occur frequently are more likely to occur in future, which should be highly vigilant in real fire management. With local SHAP, specific risk factors that vary across communities can be identified for any single community with a given period. We demonstrate the potential of this integrated machine learning scheme in improving the prediction accuracy and risk identification applicability of fire incidents, which contributes to more effective and customized fire safety management.

1 | Introduction

Community fire incidents pose a significant challenge for urban risk management. The perils associated with fire hazards in urban settings are profoundly serious. For instance, as substantiated by the data from the U.S. Fire Administration (2022), the period from 2018 to 2020 witnessed an alarming average of 1900 fatal residential fires annually in the United States, resulting in an estimated average of 2745 deaths, 625 injuries, and property losses amounting to $230 million. These statistics not only reflect the severity of fire incidents but also accentuate the necessity for comprehensive fire safety management within communities. The implementation of efficacious prevention and control strategies, coupled with meticulous inspections for potential hazards, is imperative for the preservation of human life and property (Wang et al. 2021; Palazzi et al. 2023).

The task of predicting fire incidents and identifying potential influencing factors is critically considered in the endeavor of fire safety management. Traditional methods of point fire alarms such as smoke detectors and video-based detectors can provide prompt detection and warning, but often fall short in early prediction and risk assessment (Zhang et al. 2008; Lai et al. 2010; Verstockt et al. 2013; Huang et al. 2022; Sun and Xu 2022; Festag 2023). The complexity of urban environments, as well as the dynamic nature of community activities, creates informative factors (features) that can contribute to fire risks. Constructing a predictive model that incorporates these features is vital not only for enhancing the accuracy of fire risk assessments but also for enabling communities to implement targeted prevention strategies. Therefore, by leveraging advanced predictive techniques and integrating diverse data sources, it can be better to handle fire incidents and take proactive measures to safeguard communities (Ding et al. 2020; Ouache et al. 2022).

Fire prediction has garnered increasing research attention from scholars. Despite this heightened interest, there is a paucity of studies that address a unified prediction framework for fire incidents. Some of the popular prediction models are based on time series approaches (Box and Jenkins 1968). For instance, Ferreira et al. (2020) proposed a time series-based prediction method to predict the global daily fire, using the auto-regressive integrated moving average (ARIMA) model, exponential smoothing, Prophet, and other techniques. Similarly, Ma et al. (2021) developed a seasonal ARIMA (SARIMA) model, which captures the time fluctuation trends and seasonal characteristics of data, to predict yearly fire incidents in China. Another stream of research focuses on the socio-economic determinants of fire incidents, highlighting the underlying social and economic factors that contribute to community fire risk. Fire incidents can be linked to low socio-economic status, suggesting that they are not merely accidental events but can be seen as outcomes of broader social issues (Jennings 1999; Duncanson et al. 2002; Holborn et al. 2003; Chhetri et al. 2009; Chhetri et al. 2010; Jonsson and Jaldell 2020). For example, Chhetri et al. (2009) developed models identifying socio-economic factors such as unemployment rates, the proportion of Indigenous populations, and family structures as significant predictors of fire incidents, concluding similar determinants identified by Jennings (1999). Guldåker and Hallin (2014) combined socio-economic determinants with spatio-temporal patterns to show the impact of social stressors like unemployment and overcrowding on fire risk, which coincides with the statement of Jennings (2013) that fire analysis has been accompanied with the progress of geographic information tools. Hu et al. (2019) further expanded this analysis by examining a wide range of socio-economic features, including population growth rates and industrial development levels, to predict urban fire risk. These studies collectively support the importance of considering more socio-economic driving factors in fire prediction models. In recent years, machine learning approaches have enriched model options in fire prediction, with many applied algorithms such as random forests (Madaio et al. 2016), artificial neural network and supported vector machine (Sayad et al. 2019), long short-term memory (Kumari et al. 2021), and mixed-inputs and attention mechanisms (Li et al. 2023). However, there have been few studies on the comprehensive model of fire prediction that would integrate diverse data sources, apply machine learning techniques, predict fire incidents, and understand the underlying risk factors.

Further, while models offer fire prediction results with greater accuracy, understanding the rationale behind these predictions remains a problem (Ma et al. 2022). This is particularly true for machine learning approaches that are often regarded as "black boxes" due to their opaque decision-making processes. The ability to interpret these models is crucial, not only for validating their predictions but also for gaining insights into fire risk mitigation strategies. Some methods have proposed to unravel the complexities of machine learning and provide a clearer understanding of how model inputs influence predictions, such as feature importance analysis, local interpretable model-agnostic explanations (Ribeiro et al. 2016), integrated gradient (Sun and Sundararajan 2011), and Shapley additive explanation (SHAP; see Lundberg and Lee 2017) value. Although these interpretative approaches are useful, the applications of them to fire prediction scenarios are limited.

We attempt to advance the existing literature on fire risk prediction and management in three major ways: (1) fully utilizing multi-source data including unstructured fire risk text data, which can provide crucial information to fire prediction and risk identification; (2) developing an interpretable machine learning model of fire prediction with better prediction accuracy and model interpretability to help improve the applicability; and (3) identifying most influential underlying risk factors contributing to the fire incidents from the quantitative perspective based the integrated model analysis and providing useful evidence for real fire safety management.

In this paper, we concentrate on community fires that are incidents recorded by local fire stations within specific community areas. These fires typically occur in residential and commercial buildings and are likely to be caused by factors such as unsafe electrical practices, gas cylinder mishandling, and improper storage of goods. Our study does not cover wildfires due to their distinct characteristics and data requirements. Unlike wildfires that are influenced by environmental factors such as weather and vegetation, community fires are often the result of human activities and can be mitigated through targeted interventions. By narrowing our scope to community fires, we can develop more precise models that account for the specific conditions and behaviors that lead to these incidents, leveraging data from grid-style management systems that provide detailed daily reports on communities from grid officers.

Specifically, a text dataset sourced from grid management systems is incorporated into our model, providing a detailed snapshot of community fire risk factors. We also propose to cluster the communities based on the time series information in our machine learning models, which effectively reduces the dimensionality and sparsity of the feature space that would result from using the one-hot encoding method. Consequently, the prediction performance is noteworthy; the random forest algorithm in our scheme excels in comparison to time series modeling (Prophet) and other models (logistic regression, CatBoost and XGBoost), as evidenced by its impressive F_1 score and area under the curve (AUC). These metrics support the model's precision and reliability, affirming its potential as a valuable tool for community fire prediction.

Moreover, the interpretation of machine learning in the context of fire prediction can be approached in two aspects. The feature importance analysis assesses the relative impact of each input feature on fire prediction results, while SHAP values offer a more granular interpretation by quantifying the contribution of each feature to individual predictions.

The remaining part of this paper is structured as follows. Section 2 details the highlights in the proposed scheme, encompassing multi-source information, time series clustering, Shapley additive explanation, and steps of the implementation. Section 3 delves into the specifics of the data, features, predictive models, and evaluation metrics. The results are then revealed in Section 4, providing a comprehensive analysis of the prediction and interpreting the outcomes. Finally, Section 5 wraps up the paper with a conclusion that summarizes our findings and discusses the contributions.

2 | Integrated and Interpretable Modeling

2.1 | Multi-Source Data

As the city governance become more digitalized, the use of big data on urban governance in various places has emerged, especially the unstructured text data that increasingly appears, such as the records of daily urban governance behavior (Afful-Dadzie and Afful-Dadzie 2017; Giordano et al. 2022; Liu et al. 2022; Zhang et al. 2023). For fire prediction, the performance of predictive models will be significantly improved by the integration of multi-source big data, which extends beyond mere fire event records. This approach is especially pertinent in China, where a grid-style management system, characterized by its meticulous subdivision of urban areas, provides a granular level of detail that is instrumental in identifying and categorizing potential fire hazards.

The "grid" is specially established in China as the basic governance unit that is at a more refined level than urban communities. In this paper, we innovatively incorporate text data from the grid-style management system into our fire prediction model. The text data contains the description of risk factors recorded by community grid management officers. The grid officer uses smart devices to record and enters the risks they find into the system, and the entire process of each incident resolution can be updated and tracked in real time.

By introducing such data, we extract 16 types of fire-related risks and group them into four categories: (1) from enterprises: illegal production or storage of flammable or explosive materials, unsafe industrial electricity consumption, inadequate fire extinguishing equipment as required, excessive goods stacking, lack of safety management system, hazardous chemical storage, and waste station fire risk; (2) from residents: gas cylinders placed in living rooms, illegal subdivision of rental rooms, and unsafe fire or electricity use in rental rooms; (3) from urban management: illegal constructions, construction site fire risk, and disorderly placement; (4) others: black smoke, blocked fire escape, and other risk factors. Adding these fire-related risks into the model not only refines the predictive accuracy but also tailors the fire prevention strategies to the unique characteristics of each governed area, thereby enhancing the overall effectiveness of fire safety management.

2.2 | Machine Learning With Time Series Clustering

Raw and unprocessed data, when fed directly into machine learning models, often leads to unsatisfying outcomes. This necessitates a pre-processing stage where data is curated and refined to enhance the learning efficiency. For fire prediction where data is inherently sequential, it is crucial to capture the patterns of fire incidents and extract more temporal information before training the model. For example, the traditional one-hot encoding method for the community label information will naturally result in high dimensionality and sparsity of the feature space, which further affects the prediction accuracy and make the model interpretability more complicated. By employing time series clustering algorithms, we can develop a more concentrated set of features from the time series, thereby increasing the accuracy of machine learning in fire prediction.

In this paper, the dynamic time warping (DTW) method is used to calculate the similarities between different time series of fire incidents, as shown in Algorithm 1. DTW aligns two time series by finding a matching between their time indexes that minimizes the Euclidean distance between them, accommodating time shifts and enabling a comparison of temporal sequences (Bellman and Kalaba 1959; Sakoe and Chiba 1978). This alignment helps in generating clustered features that are more concentrated and informative.

The process involves two main modules: (1) Alignment path. DTW seeks a path that minimizes the total distance between corresponding points in two time series, ensuring the path starts and ends at the corners of the matrix and moves forward without jumps. This alignment captures the temporal dynamics of the sequences. (2) Cumulative distance matrix. To find the optimal alignment path efficiently, DTW uses dynamic programming to construct a cumulative distance matrix. Each element in this matrix contains the distance of the current element plus the minimum distance of its neighboring elements, allowing for the determination of the similarity between the two sequences.

ALGORITHM 1 | Dynamic time warping algorithm.

```
n ←|X|
m ←|Y|
dtw[] ← new [n×m]
dtw(0, 0) ← 0
for i = 1; i ≤ n; j ++ do
    dtw(i, 1) ← dtw(i-1, 1)+d(i, 1)
end for
for j = 1; j ≤ m; j ++ do
    dtw(1, j) ← dtw(1, j-1)+d(1, j)
end for
for i = 1; i ≤ n; j ++ do
    for j=1; j ≤ m; j ++ do
        dtw(i, j) ← d(i, j)+min{dtw(i-1, j); dtw(i, j-1); dtw(i-1, j-1)}
    end for
end for
return dtw
```

By employing DTW and clustering the time series data, the approach aims to reduce the dimensionality and sparsity of the feature space, thereby improving the prediction accuracy and interpretability of the machine learning models used for fire prediction. This method ensures that the temporal information inherent in the data is effectively captured and utilized in the predictive modeling process.

2.3 | Shapley Additive Explanation

SHAP is proposed to enhance the interpretability of machine learning, which is grounded in game theory and offers a systematic approach to quantify the impact of each feature on the prediction. The advantage of SHAP in this paper lies in its ability to provide a detailed decomposition of the fire prediction into the individual contributions of each risk factor, thereby revealing not only the importance of a risk factor but also its positive or negative effect on the fire prediction outcome.

Given the set of risk factors $\mathbf{F}=\{1, 2, ..., p\}$, all risk factors in a certain subset $S \subseteq \mathbf{F}$ cooperate towards the fire prediction outcome $f(S)$. The Shapley values redistribute the total fire prediction outcome value $f(\mathbf{F})$ among all risk factors based on their average marginal contribution across all possible S. More specifically, the risk factor j's marginal contribution with respect to S is

$$\psi_j(S) = f(S \cup \{j\}) - f(S) \tag{1}$$

Then, the corresponding Shapley value ψ_j measures the risk factor j's contribution based on the average across all $S \subseteq \mathbf{F} \backslash \{j\}$:

$$\psi_j = \sum_{S \subseteq \boldsymbol{F} \backslash j} \frac{|S|!(p-|S|-1)!}{p!} \cdot \psi_j(S), \tag{2}$$

which is also called the global SHAP value of the risk factor j. Here, the coefficient $\frac{|S|!(p-|S|-1)!}{p!}$ is used as normalization term based on the number of choices for the subset S.

Denote $x^{(i)}$ as the ith observation of the structured data collected from multi-sources of fire incidents and grid-style management systems, and $C=\mathbf{F}\backslash\{S\}$, then the SHAP value of the risk factor j for the ith observation can be calculated as follows:

$$\psi_j^{(i)} = \sum_{S \subseteq \mathbf{F} \backslash j} \frac{|S|!(p-|S|-1)!}{p!} \left(\int f\left(x_{S \cup j}^{(i)} \cup X_{C \backslash j}\right) d\mathbb{P}_{X_{C \backslash j}} - \int f\left(x_S^{(i)} \cup X_C\right) d\mathbb{P}_{X_C} \right), \tag{3}$$

which is also called the local SHAP value of the risk factor j for the ith observation. The global SHAP value ψ_j can be obtained by averaging over all the corresponding local SHAP values for the observations accordingly.

We adopt both the global SHAP and local SHAP adaptively to analyze the contributions from different risk categories to the fire prediction, including both the overall level and observation level, accompanied with special attention to the comparison of SHAP results between risky and safe communities. To make full use of fire related information, features in addition to risk factors in Table 2 are also incorporated into the input features $\mathbf{F}$. In this way, SHAP is selected to identify the fire risk in detail and to interpret the black-box machine learning in the context of fire prediction.

2.4 | Steps of the Implementation

The structured approach that ensures a methodical analysis of fire prediction and risk factors, as outlined in Figure 1, is as follows:

Step 1. Preprocessing the community fire data for prediction and risk factor identification. A text processing method is adopted to convert the unstructured text data into regular categorical variables and merge them with the structured data to form the feature space.

Step 2. Instead of adopting one-hot encoding for community labels, creating new features with time series clustering methods for communities based on their time series of fire incidents and risk factors. Also performing down-sampling on the data to address the issue of class imbalance and to improve the prediction performance.

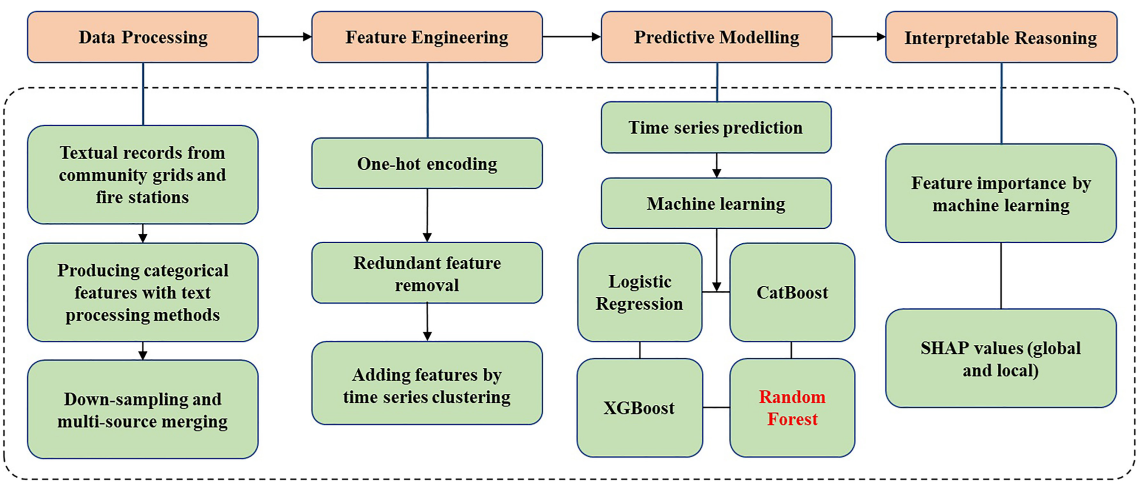

FIGURE 1 | The flow chart of fire prediction and risk specification.

Step 3. Performing candidate machine learning methods, such as logistic regression, XGBoost, CatBoost, and random forest, to predict the fire incidents.

Step 4. Employing model interpretability methods, including feature importance analysis and SHAP value analysis (both global SHAP and local SHAP), to examine the importance of features and provide insights for fire prevention.

3 | Experimental Design

3.1 | Data

The data source described in the scheme is a collection of government big data from an anonymous district in Guangdong Province, China. The local government aims to enhance urban management and governance through the construction of extensive databases. These databases encompass a wide range of topics, including demographics, housing, legal matters, and spatial geography. To support model training and academic research, a desensitized sample database was created by the local government, providing real data for the analysis of fire incidents and community issues within the district.

Figure 2 demonstrates the desensitized sample data and how we proceed with it. The upper side of the figure comes from the grid-style management system, which contains descriptions of risk factors recorded and updated by grid officers. For example, if we want to make a prediction on February 10, 2020, we search the grid-style management system for risks observed by grid officers over the past 7 days. Some of these risks are handled immediately and labeled as "resolved." However, some risks remain unresolved and potentially lead to fire incidents. We focus on these unresolved risks that are classified into 16 types based on their text descriptions. These records are then aggregated to present the number of unresolved instances for each risk type, showing the potential risks over the past 7 days as of February 10, 2020.

Simultaneously, we collect information from the lower side of Figure 2, representing emergency alarm reports. These reports contain emergency calls from residents to fire stations. Upon receiving these calls, fire stations dispatch rescue teams to the community. We focus on true emergency alarms labeled as confirmed, extracting calls that mention "fire" or "smoke" as fire incidents. To avoid wrong calls, we also include incidents that are identified as fire related from the table of rescue records. These combined incidents are aggregated to evaluate whether there are fire incidents in the next 7 days since February 10, 2020. We map the upper and lower parts of Figure 2 according to the relationship between fire stations and the communities they serve. In the merged structured table, for predictions on February 10, 2020, we have features representing unresolved risks over the past 7 days and the occurrence of fire incidents within the next 7 days. With additional features in Table 2, we train machine learning models to learn the relationship between fire risks and fire incident probabilities based on this merged table.

The data trimmed from the desensitized database cover 111 communities in the district and span from January 1, 2018, to January 31, 2021. The grid-style management text data used in our study are collected from a comprehensive system that

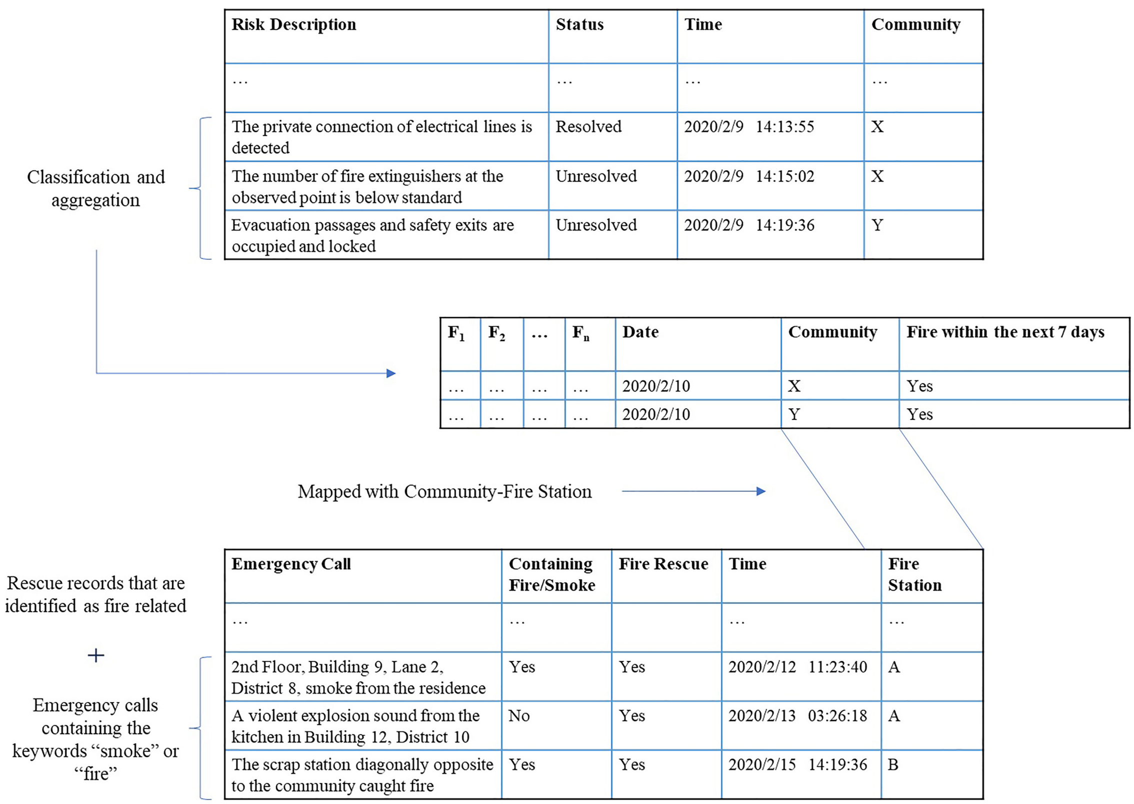

FIGURE 2 | Demonstration of the desensitized sample data.

assigns grid officers to report daily on risk factors within their designated areas. This ensures that the data cover the entire research area, providing a robust foundation for our analysis. Table 1 presents the descriptive statistics of these signals for the whole period of 1127 days. To avoid underestimating the number of fire incidents, we create a target variable by taking the union of these signals. This target variable captures the occurrence of any fire-related event in a given community within the next 7 days.

However, even we have merged these fire records, the data set is still imbalanced, as only 25.87% of the samples belong to the positive class. To address this issue, we apply a down-sampling technique that randomly selects a subset of the negative class to match the size of the positive class (Freeman et al. 2012). After down-sampling, we obtain a balanced data set with 65,858 samples, where each class has a 50% proportion.

3.2 | Features

We only consider China in this paper because China's adoption of the recently established grid-style management system presents a unique opportunity for more informative features in machine learning. The decision to focus on the grid-style management system for fire prediction is strategic; it allows for the exploration of the effectiveness of this grid-style management in urban governance. The system's coverage and daily reporting capabilities is expected to make it an ideal source of granular data that can significantly enhance the prediction accuracy. For each type of community risk in the grid-style management system, we calculate the number of cases that were registered but not closed in the previous 7 days (excluding the current day) based on the preprocessed community risk data, as revealed in Table 2.

Besides the risk categories, we also use historical fire records, location, time, and clusters as features. Historical fire records indicate whether there were smoke, fire police rescue, or fire events in each community in the past 30 days. Location is the community to which the fire record belongs, encoded using one-hot encoding (clustering later). Time is the day, month, and year information, and our data covers the research period of fire records. Clusters include community clusters and risk factor clusters; they are determined by historical fire incident frequency and historical risk factor distribution respectively. One might suggest that meteorological data such as temperature, humidity, and wind could also be useful predictors for the fire prediction model. However, since our study only focuses on communities within a single district, the weather variations among different communities are negligible. Therefore, we do not consider the meteorological data in our model for now.

The original data set includes categorical dummies for the 111 communities to which the samples belonged. However, this results in a high-dimensional and sparse feature space, which will affect the performance of the models. To reduce the complexity and leverage the community information more effectively, we apply clustering algorithms to group the communities into clusters and use the cluster labels as new features instead of the community dummies. Using the DTW distance, we determine that the optimal number of clusters is four. Figure 3 shows the temporal patterns of the four cluster centroids, which are clearly distinct.

3.3 | Predictive Models

To validate the practical effectiveness of our scheme, we adopt four mainstream machine learning prediction models in our analysis:

- Logistic regression. This model is characterized by its use of the sigmoid function to transform continuous values into binary outcomes. It operates under the assumption that the data is linearly separable and focuses on estimating the coefficients that influence the log odds of the outcome variable.
- XGBoost (extreme gradient boosting). It stands out for incorporating second-order derivative information when optimizing the loss function, which is a departure from the traditional gradient boosting decision tree method (Chen and Guestrin 2016). It builds decision trees sequentially, with each tree learning from the errors of the previous ones.
- CatBoost (categorical boosting): This model is recognized for its efficiency in handling categorical features. Like XGBoost, it is a boosting method that relies on decision trees but is specifically optimized for categorical data (Prokhorenkova et al. 2018).
- Random forest. This model employs an ensemble learning strategy that combines the predictions of multiple decision trees (Breiman 2001). By averaging the results, it reduces the risk of overfitting and improves the accuracy of the predictions.

The selection of algorithms for developing the fire prediction model is based on two key criteria: adaptability to the binary characteristic of the time series dependent variable prediction and compatibility with interpretable machine learning tools like

TABLE 1 | Summary statistics of the target variable and related signals.

	Police_FUTURE_7	Smoke_FUTURE_7	Fire_FUTURE_7	Target
Count	125,097	125,097	125,097	125,097
Mean	0.1900	0.0720	0.1679	0.2587
Std	0.3923	0.2584	0.3738	0.4379

TABLE 2 | Description of features.

Predictors	Description
Increased short-term cases in the past 7 days (excluding that day)	
Illegal production INC_7	Illegal production or storage of flammable or explosive materials
Unsafe power INC_7	Unsafe industrial electricity consumption
Lacking equipment INC_7	Inadequate fire extinguishing equipment as required
Excessive stacking INC_7	Excessive goods stacking
Lacking management INC_7	Lack of safety management system
Chemical storage INC_7	Hazardous chemical storage
Waste hazard INC_7	Waste station fire risk
Gas cylinders INC_7	Gas cylinders placed in living rooms
Room split INC_7	Illegal subdivision of rental rooms
Unsafe rental INC_7	Unsafe fire or electricity use in rental rooms
Illegal construction INC_7	Illegal construction
Construction hazard INC_7	Construction site fire risk
Disorderly placement INC_7	Disorderly placement
Black smoke INC_7	Black smoke
Blocked escape INC_7	Blocked fire escape
Others INC_7	Other risk factors
Increased long-term cases in the past 30 days (excluding that day)	
Police_PAST_30	Whether there was a fire police rescue event during the past 30 days
Smoke_PAST_30	Whether there was a smoke alarm report during the past 30 days
Fire_PAST_30	Whether there was a fire alarm report during the past 30 days
Increased long-term cases in the prediction day	
Police	Whether there was a fire police rescue event on the prediction day
Smoke	Whether there was a smoke alarm report on the prediction day
Fire	Whether there was a fire alarm report on the prediction day
Time, location, and clusters	
Day	Indicators of the day
Month	Indicators of the month
Year	Indicators of the year
Weekday	Indicators of whether the day is a weekday or weekend
Community	Indicators of the community location
Cluster_fire	Indicators of the community clusters (by historical fire incident frequency)
Cluster_hazard	Indicators of the risk factor clusters (by historical risk factors distribution)

SHAP. This leads to the choice of logistic regression, XGBoost, CatBoost, and random forest.

Logistic regression is widely used in risk prediction for its good performance, and as one kind of generalized linear model, it performs feature sparsity when combined with the lasso penalty, which can provide good explainability in applications and serves as a good comparison choice. XGBoost generally achieves extraordinary performance in prediction tasks (see Bindingsbø et al. 2023). Similar to the XGBoost, Catboost is recognized for not only its prominent prediction accuracy but also its efficiency in handling categorical features. Both of these two methods show good compatibility with SHAP, so we choose them within the category of boosting models. By ensembling the predictions of multiple decision trees, the random forest model shows not only good prediction performance with less overfitting but also good interpretability as tree-based models.

Other models have been considered, but they are not chosen for specific reasons. Ridge regression, although useful for handling multicollinearity, does not offer the same level of feature sparsity and interpretability as lasso penalty when combined with logistic regression. Decision trees, despite their interpretability, can be prone to overfitting and may not perform as well as ensemble methods like random forest. AdaBoost, although a strong boosting method, does not match the performance and efficiency of XGBoost and CatBoost, especially in handling categorical feature. Deep learning models like FNN and LSTM, while powerful, may not align as well with the interpretability requirement and tend to require more data resources.

For the baseline, common methods used for time series prediction include classical autoregressive integrated moving average (ARIMA), exponential smoothing methods, and Holt-Winters. However, under current fire-prediction setting, these methods are not suitable for the binary characteristic of the time series dependent variable. Instead, we include the Prophet model, to provide a baseline prediction result of time series (Taylor and Letham 2018). Building on the foundation of the regression, the Prophet model is established by Facebook as an automated approach that utilizes the generalized additive model (Hastie and Tibshirani 1990), which is robust to missing values and structural change. It handles outliers well and achieves good performance in various time series prediction tasks.

To tune the hyperparameters of the models, we use a cross-validation technique to optimize them. We use grid search method for the best hyperparameters selection within all the possible hyperparameters combination sets, as shown in Table 3. All other parameters not mentioned take the default values. Since our data set is a time series, we employ a time-series k-fold cross-validation algorithm to split the data into training, validation, and testing sets. This algorithm splits the time series

TABLE 3 | Parameter settings adopted in the machine learning models.

Parameter	Value
Logistic regression	
Penalty	‘l1’
XGBoost	
n_estimators	[100, 150, 200, 300]
learning_rate	[0.01, 0.015, 0.025, 0.05, 0.1]
gamma	[0, 0.05, 0.1, 0.3, 0.5, 0.7, 0.9, 1]
max_depth	[3, 5–7, 9, 12]
min_child_weight	[1, 3, 5, 7]
CatBoost	
depth	[3, 5–7, 9]
iterations	[200, 400, 600, 800]
learning_rate	[0.01, 0.015, 0.025, 0.05, 0.1]
Random forest	
n_estimators	[100, 150, 200, 300]
max_depth	[4–8]
min_samples_leaf	[1–3]
criterion	[‘gini’, ‘entropy’]

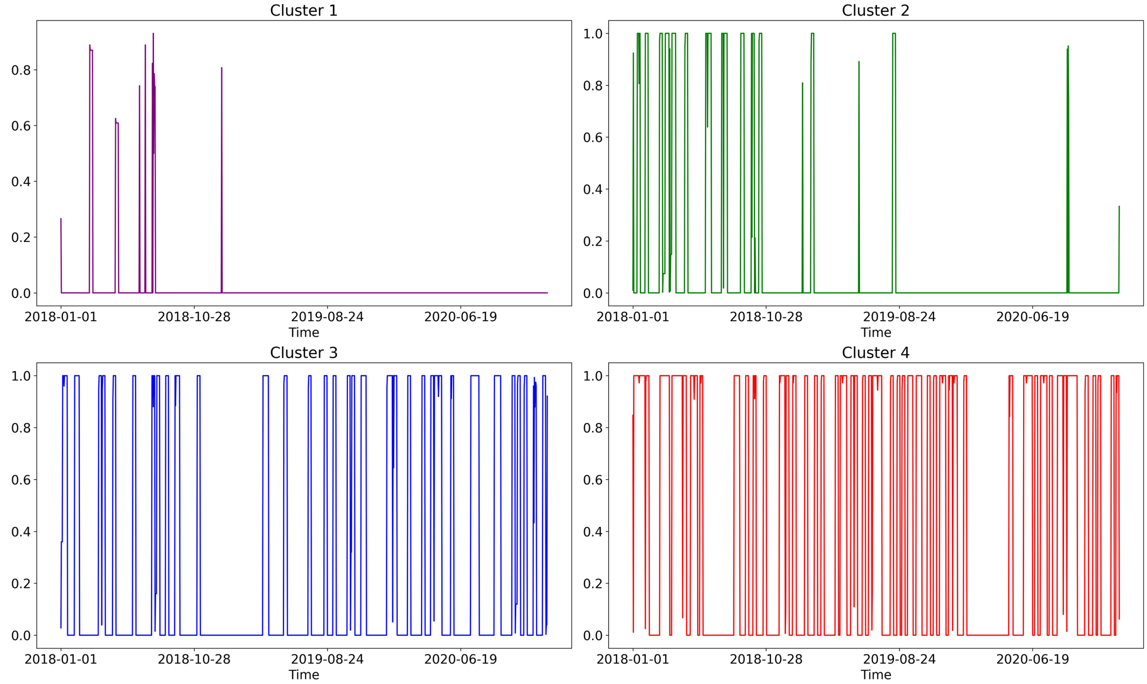

FIGURE 3 | The distribution of overall community fire accidents with respect to time according to the “Cluster_fire” variable.

data, which are observed at fixed time intervals, into train/test sets in a sequential order. In each split, the test set has higher indices than the previous ones. That is, the algorithm uses the first k folds as the training set and the $(k+1)$-th fold as the test set, as shown in Figure 4. We choose $k=5$ to split the data into three subsets: a training set with 52,015 samples from January 1, 2018, to July 5, 2020; a validation set with 10,402 samples from July 6, 2020, to December 31, 2020; and a testing set with 3441 samples from January 1, 2021, to January 1, 2021.

3.4 | Evaluation Metrics

Consider that a true positive (TP) observation occurs when the model accurately predicts a positive class outcome (corresponding to the data point with fire event), while a true negative (TN) occurs when it accurately predicts a negative class outcome (corresponding to the data point without fire event). Conversely, a false positive (FP) and a false negative (FN) occur with erroneous predictions. Based on TP, TN, FP, and FN, *accuracy* measures the overall proportion of correct predictions, with true predictions divided by all predictions. *Precision* assesses the reliability of positive predictions with $\frac{TP}{TP+FP}$, while recall evaluates the coverage of positive outcomes with $\frac{TP}{TP+FN}$. F_1 *score* combines *precision* and *recall* into a single metric that reflects a balanced performance by $\frac{2 \cdot Precision \cdot Reall}{Precision + Recall}$. Since the class labels are heavily unbalanced for real fire event data, even if all the unknown samples are directly predicted as positive without any modeling, a high prediction accuracy measured by $\frac{TP+TN}{TP+TN+FP+FN}$ will still be achieved. So instead of using accuracy as an evaluation metric, we mainly focus on the *precision*, *recall*, and F_1 *score*.

4 | Results and Discussion

4.1 | Prediction Performance

The prediction results are presented in Table 4. We compare the use of community dummies and time series clustering to highlight the difference between them. Community clusters and risk factor clusters determined by the DTW distance enhance prediction performance on F1-score and AUC with significantly fewer features. Overall, the proposed machine learning models with time series clustering show excellent performance compared to traditional machine learning methods.

Additionally, the four machine learning methods outperform the baseline Prophet model, showcasing the benefits of machine learning in utilizing information from various features over single variable time-series prediction. Upon comparing the four machine learning methods for clustered features, it is observed that the random forest model excels over the other three in most

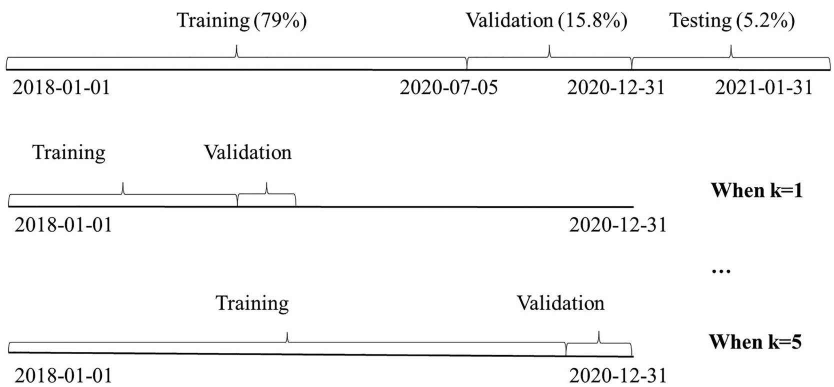

FIGURE 4 | Splits in the time-series k-fold cross-validation.

TABLE 4 | Prediction results of the four proposed models.

Setting	Model	Precision	Recall	F1-score	AUC
Baseline	Prophet	0.506	0.622	0.560	0.661
Traditional machine learning methods (one-hot community feature)	Logistic regression	0.556	0.709	0.627	0.713
	XGBoost	0.506	0.825	0.626	0.710
	CatBoost	0.497	0.831	0.623	0.705
	Random forest	0.509	0.816	0.626	0.709
Revised machine learning with time series clustering features	Logistic regression	0.544	0.762	0.637	0.722
	XGBoost	0.543	0.818	0.653	0.735
	CatBoost	0.546	0.820	0.656	0.736
	Random forest	0.543	0.824	0.656	0.737

Note: Besides dummies or clusters, the features in Table 2, including text-based features, location, and time, are all adopted in the models, though they are not listed here for simplicity.

metrics. With community clusters and risk factor clusters as features, the random forest achieves a precision of 0.543, a recall of 0.824, an F1-score of 0.656, and an AUC of 0.737.

Using these clustered features, our model outperforms particularly in the recall score, which reflects the model's ability to correctly find fire incidents as many as possible. For example, the Firebird project, as mentioned above in Madaio et al. (2016), used machine learning and achieved a recall score of 0.714, while our random forest has a significantly higher recall score 0.824. Since predicting positive cases (when a fire incident occurs) is more important than predicting negative cases in fire incident prediction, the model with higher recall rates is more effective in fire prediction.

The improvements in our model's performance, as shown in Table 4, may appear modest. This is because that our baseline model already incorporates multi-source data, which sets a high standard for comparison. Therefore, even a 0.05 improvement in the precision metrics represents a significant enhancement in predictive accuracy, given the context in which the performance is achieved in the baseline. Moreover, these models' performance metrics, such as AUC at 0.737 and recall at 0.824, indicate an excellent predictive capability considering the uncertainty of fire incidents in our prediction settings. Predicting fire incidents for a microlevel and a short period, such as within certain community areas in a week, is inherently challenging due to the random nature of such events. Despite this, these models demonstrate a reliable ability to forecast these incidents for a specific community during a given period, which is important for effective fire risk management. In addition, beyond prediction accuracy, our approach emphasizes the identification of risk factors, which aims at providing feasible guidance for the decision-making of real community fire risk management in advance. This dual focus aims not only to predict fire incidents but also to provide actionable insights into the underlying causes, thereby addressing both prediction and practical application needs.

4.2 | Relative Importance of Features

The feature importance analysis of the random forest reveals the most influential factors for the outcome. Clustering information stands out as the most crucial feature for prediction, as depicted in Figure 5. This is followed by historical records, specifically Police_PAST_30, Fire_PAST_30, and Smoke_PAST_30, along with the Others_INC_7 feature. Time information ("Month") also plays a significant role. The random forest's feature importance aligns well with other models, with only minor differences. All four models have some common important features, including cluster information and historical records (Police_PAST_30, Smoke_PAST_30, Fire_PAST_30). However, logistic regression gives more weight to other risk factors such unsafe rental (unsafe fire or electricity use in rental rooms) and excessive stacking (excessive goods stacking).

4.3 | Interpretation With Global SHAP

SHAP values provide an interpretation of machine learning by quantifying the impact of a feature's presence on the prediction probability of fire incidents. Figure 6 shows the average SHAP value for each feature, with Cluster_fire being the most influential, followed by historical records and month information. This aligns with the previous feature importance analysis from the random forest model. Further, Figure 7 details the prediction for each data point and how each predictor influences the outcome. A higher Cluster_fire value increases the likelihood

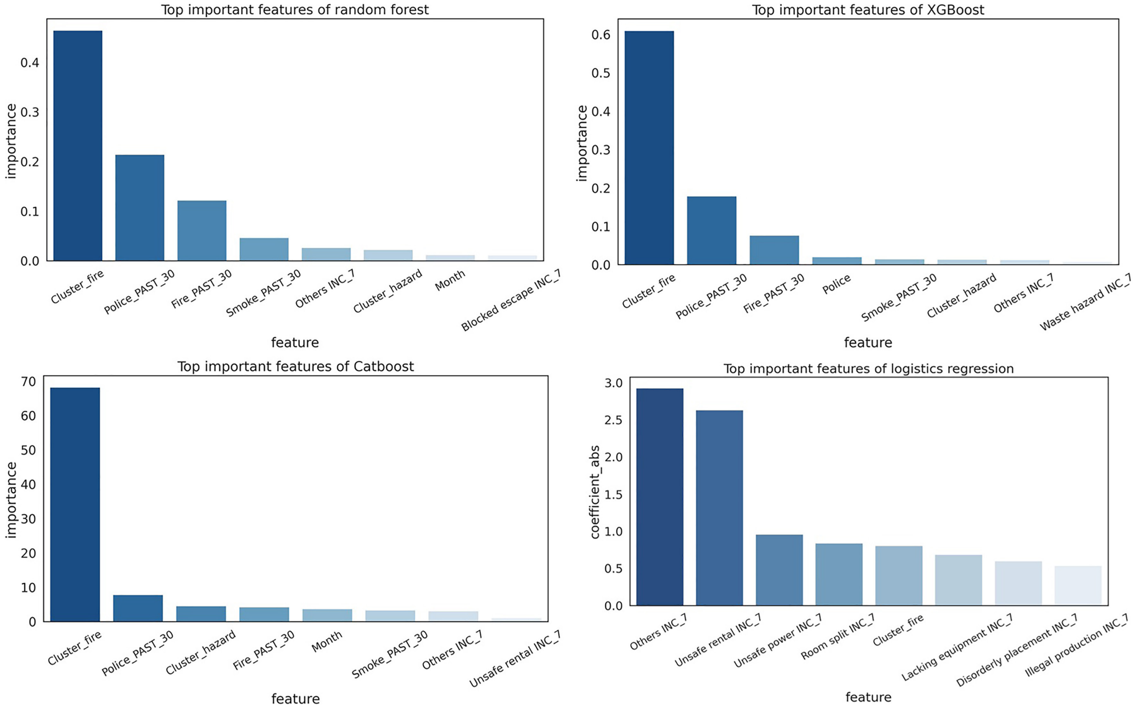

FIGURE 5 | The feature importance of proposed models.

of a positive class prediction, correlating with a greater probability of fire incidents. The dummies containing historical records also affect the prediction direction based on their values, where 1 suggests a positive influence and 0 a negative one.

Additionally, the distribution of key predictors on the SHAP value is examined through scatter plots in Figure 8. These plots confirm the significant impact of the selected predictors on the predictions, consistent with earlier figures. The month feature exhibits a pattern in its influence, where values near 0 and 1 negatively affect the prediction, while mid-range values have a positive effect, indicating higher fire risk during summer months (July and August), corroborating the exploratory data analysis findings.

4.4 | Interpretation With Local SHAP

The local SHAP value analysis aims to understand the impact of each risk factor on fire probability and to examine individual incidents for potential intervention. Figure 9 exemplifies this with two observations: one positive and one negative. The positive observation's SHAP value of 0.33 indicates a higher than 0.5 probability of a fire incident after sigmoid transformation. Influential features for this prediction include Smoke_PAST_30, Fire_PAST_30, Cluster_fire, and Cluster_hazard, which all have positive values, aligning with their respective meaning known for higher fire incidence. Conversely, Police_PAST_30 has a negative impact, with a value of 0 indicating no fire police rescue in the past 30 days.

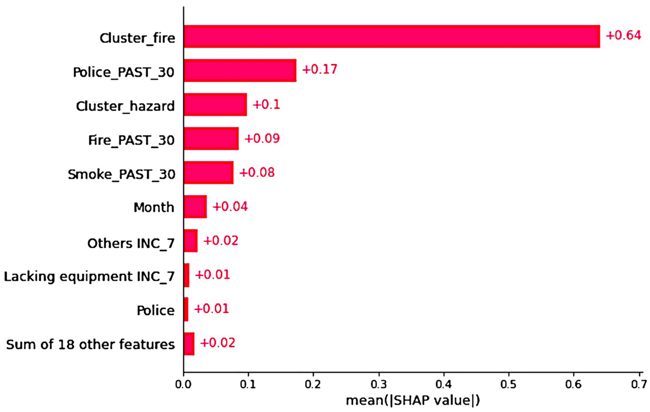

FIGURE 6 | The mean SHAP value of each predictor.

To further investigate the local SHAP, we select an anonymous Community X, a typical high-risk community that belongs to the highest level of fire clusters and fire hazard clusters. The distribution of key features of Community X is compared to their SHAP values in Figure 10. The blue curve represents the actual distributions of the features, and the orange line represents the distributions of the SHAP values of the features for this specific community. First, we examine the Others INC_7 and Lacking equipment INC_7. It can be seen that when the feature values peak (indicating higher risk), their SHAP values also increase, suggesting a positive influence on the prediction outcome. Next, for Police_PAST_30, a most consistent value of 1 results in positive SHAP values around 0.16. Interestingly, an absence of fire police rescue in April 2020, indicated by a value of 0, causes the SHAP value to drop to −0.2, negatively influencing the prediction. This pattern is observed with other features like Fire_PAST_30, Smoke_PAST_30, and Month. Lastly, the clustered features Cluster_fire and Cluster_hazard are assessed for community X. The magnitude of these SHAP values, around 0.5, is significantly larger than those for other features, highlighting the dominant influence of clustered features on the prediction outcome.

4.5 | Comparisons Between Communities

Following the analysis of Community X, the comparison extends to all communities in Cluster 4, the highest level in fire clusters

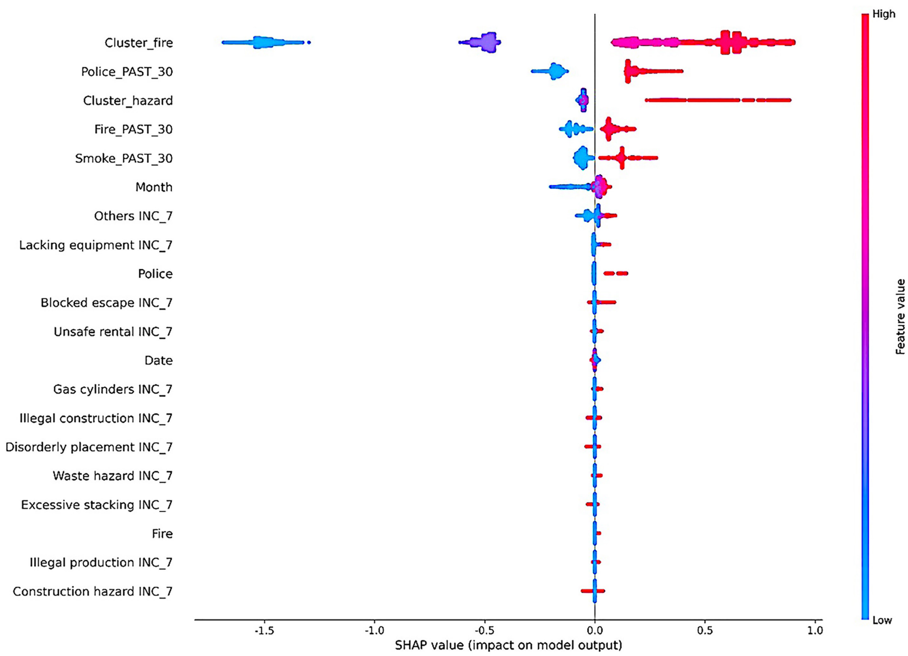

FIGURE 7 | The summary plot of the impact of each predictor's direction on the prediction.

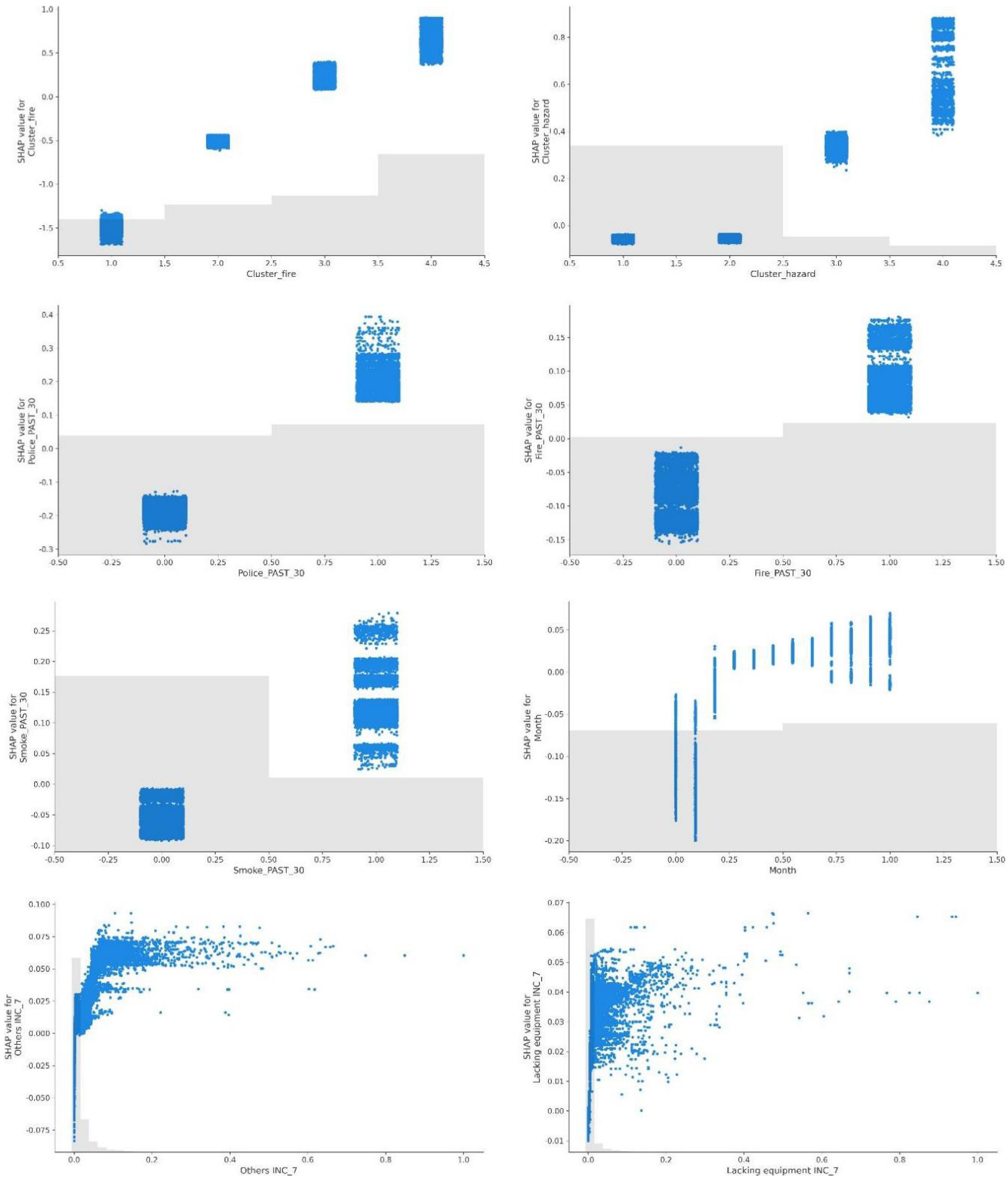

FIGURE 8 | The scatter plots of main predictors' values and their SHAP values.

FIGURE 9 | Local SHAP value of two randomly chosen incidents.

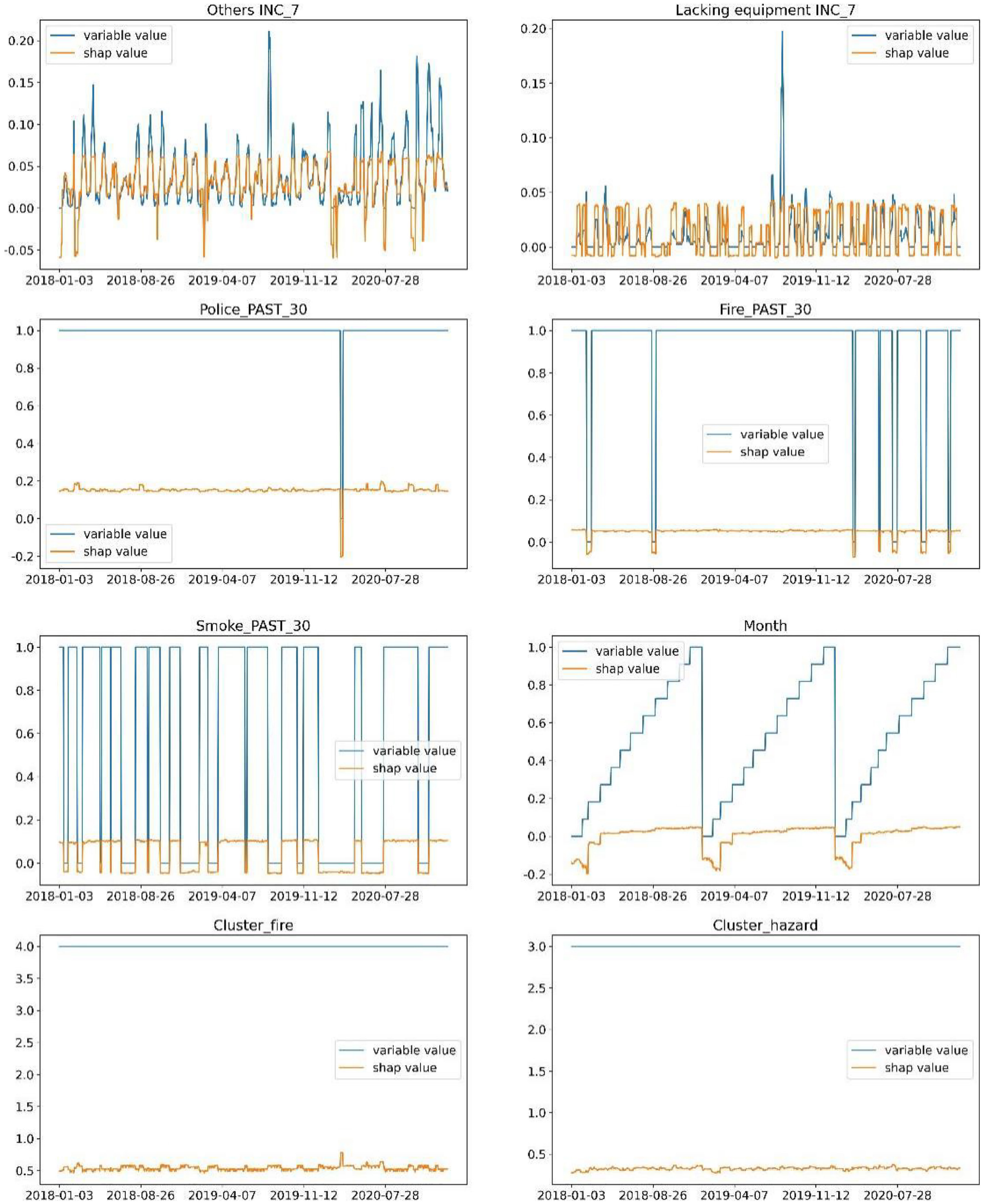

FIGURE 10 | SHAP value comparison for Community X: main features.

determined by the DTW distance. As shown in Figure 11, by averaging the values across these communities, we once again confirm that the feature distributions align with their SHAP values. Notably, the Fire_PAST_30 feature demonstrates a consistent pattern; its SHAP value turns negative as the feature's value decreases. The Cluster_fire feature, consistently at 4 for these communities due to Cluster 4, shows SHAP values around 0.66, indicating a positive fire probability influence.

We also explore the communities in Cluster 1 (the lowest level in fire clusters), which have a low probability of fire incidents. Figure 12 shows that the distribution of features and their SHAP values are still similar. However, the different part is that the values of the historical records are much lower for Cluster 1 than for Cluster 4, and the SHAP values of these features are mostly negative. For instance, Smoke_PAST_30 always negatively impacts predictions, which may contribute to lower fire probability results for Cluster 1. The Cluster_fire feature for Cluster 1 communities has SHAP values around −1.6, signifying a substantial negative probability influence.

Based on the analysis above, we can see that the Cluster_fire feature has a strong influence. The mean SHAP value for it across different clusters further illustrates this point; it is positive for

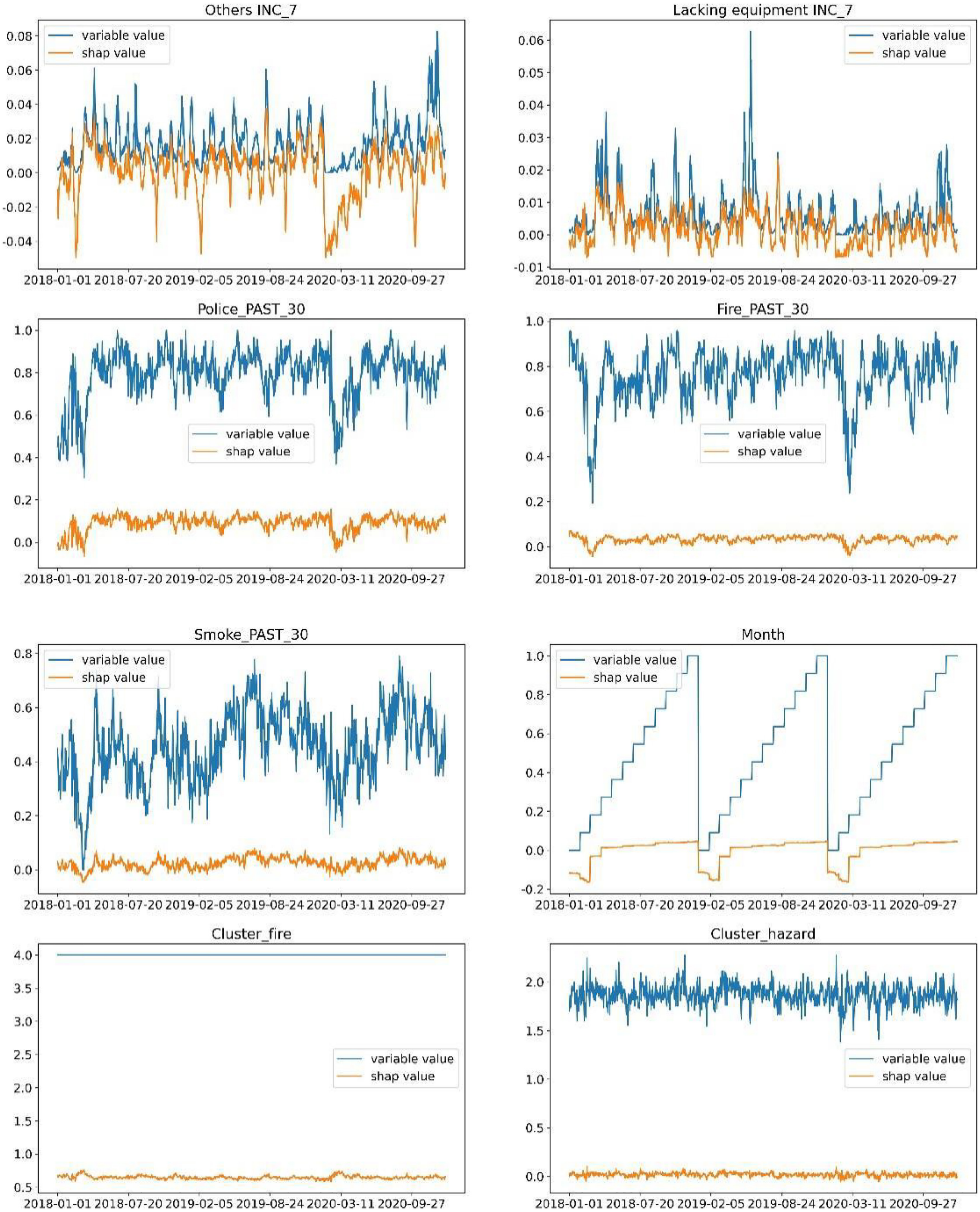

FIGURE 11 | SHAP value comparisons for high-risk communities.

communities in higher-risk clusters (Clusters 3 and 4) and negative for those in lower-risk clusters (Clusters 1 and 2), with a magnitude much greater than other features. The local SHAP analysis validates the importance of community clusters, as well as historical records, fire hazards, and month variables in predictions, with Cluster_fire emerging as the most critical predictor.

4.6 | Threats to Validity

Despite of the novelty to addressing the problems in fire safety management, this paper has certain limitations about study area, modeling techniques, and causality that offer avenues for further enhancement. For study area, this paper's scope is confined to communities in China, which may limit the generalizability of the findings. Fire incidents and their predictors can vary across different regions due to local factors. Expanding the research to include data from multiple countries can enhance the model's applicability and robustness. For modeling techniques, the proposed scheme does not employ neural network or deep learning techniques due to their black-box characteristics (Pekel Ozmen and Ozcan 2022). However, these more advanced modeling approaches, such as the long short-term memory model by Hochreiter and Schmidhuber (1997),

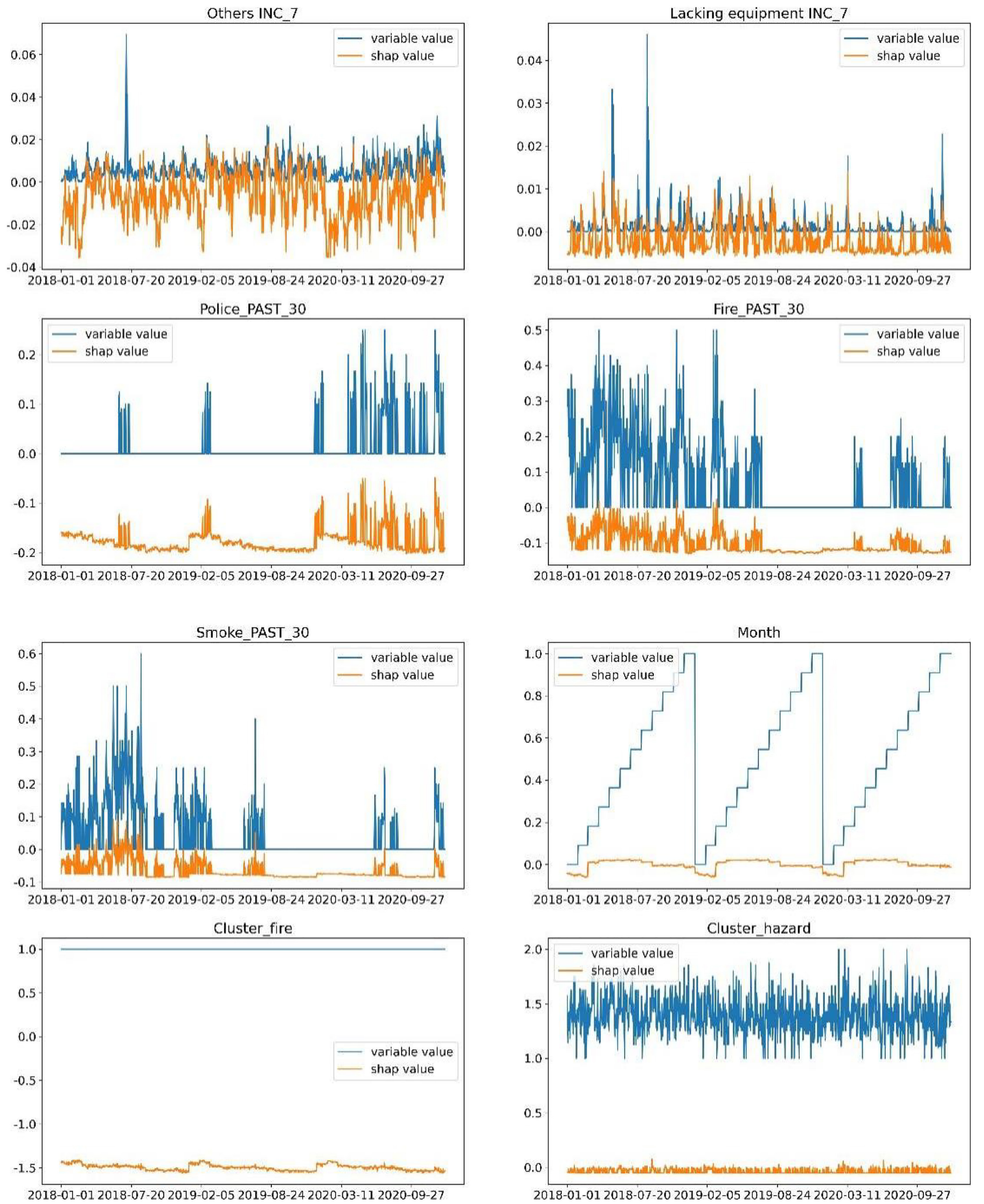

FIGURE 12 | SHAP value comparisons for low-risk communities.

may potentially capture the temporal features of fire incidents more effectively than traditional machine learning methods and further improve the prediction accuracy. For causality, this paper focuses on predictive accuracy and interpretability but does not incorporate formal causal inference methods (Rubin 1974; Pearl 1995). While feature importance and SHAP values provide insights into the factors influencing fire risk, future research can benefit from applying causal inference methods to better understand the underlying causes and individual treatment effects.

5 | Conclusion

In this paper, we propose a novel machine learning scheme aimed at enhancing the prediction of fire incidents. This proposed scheme is not merely a predictive tool; it is an integrated system that combines diverse data sources, time series clustering techniques, machine learning, and interpretable reasoning. In the comparison with Prophet that is merely based on time series data, machine learning models (Logistic Regression, XGBoost, CatBoost and Random Forest) in our proposed scheme stand out for predictive capabilities, as evidenced by impressive F1-score and AUC metrics. The performance of random forest is particularly noteworthy due the importance of higher recall rates in identifying fire incidents. In terms of interpretability, our scheme employs SHAP values to elucidate the prediction, offering transparent insights into the factors influencing fire incidents. Among various features, community clusters determined by the DTW distance, as well as historical records on fire police rescue, smoke alarms, and fire alarms, emerge as significant predictors.

Our work makes several contributions to the field of fire risk prediction and management. First, this paper introduces an innovative machine learning scheme that integrates unstructured text data from grid management systems, time series clustering adaptive to features and model interpretation mechanisms to machine learning. Existing studies on point fire alarms, time series prediction and machine learning methods are limited in their respective aspects on early prediction, data integration and opaque decision-making processes. Our integration allows for a more comprehensive analysis of community fire risks. Second, this paper enhances the interpretability of machine learning models in the context of fire prediction. It employs feature importance and SHAP values to provide a transparent view of the factors influencing predictions. This approach not only validates the models predictions but also offers insights into potential fire risk mitigation strategies. The ability to interpret the model's outputs is valuable, as it enables us to understand the rationale behind the predictions and to trust the model's recommendations. Third, this paper identifies community clusters and historical records as the most influential feature in predicting fire incidents. It supports the capability to discern and prioritize the most critical predictors of fire incidents. By pinpointing these key factors, this paper provides safety measures that can be used to focus fire prevention efforts where they are most needed.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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